# Test Markdown

**TL;DR**
Text für TL;DR …
In Parr (2014) wird geschrieben, blablablabla …
Wir können hier sowohl Inline-Math ($a^i b^{2*i}$) als auch Block-Math:

$$\Phi(\mathbf{g}_i) = F(\Gamma(\mathbf{g}_i)) - w \cdot \sum_j (Z_j(\Gamma(\mathbf{g}_i)))^2$$

Code sollte auch gehen: `inline`, aber auch block:

```
fun fib(x) {
    if (x == 0) {
        return 0;
    } else {
        if (x == 1) {
            return 1;
        } else {
            fib(x - 1) + fib(x - 2);
        }
    }
}

var wuppie = fib(4);
```

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Convert Markdown sources to GFM passing

## Hello World

Hier ist normaler Markdown-Text, mit **fett** und auch *kursiv*.

- Stichpunkt 1
- Stichpunkt 2
- Stichpunkt 3

1. Aufzählung 1
2. Aufzählung 2
3. Aufzählung 3
    1. Unterpunkt 3.1
    2. Unterpunkt 3.2

Hier die Pandoc-Markdown mark-Erweiterung.

## Math

### Inline

$\mathbf{g} = (g_1, \ldots, g_m) \in \{0, 1\}^m$

- $a^i b^{2*i}$ ist nicht regulär
- $a^i b^{2*i}$ für $0 \leq i \leq 3$ ist regulär

### Block

$$\Phi(\mathbf{g}_i) = F(\Gamma(\mathbf{g}_i)) - w \cdot \sum_j (Z_j(\Gamma(\mathbf{g}_i)))^2$$

$$p_{sel}(\mathbf{g}_k) = \frac{\Phi(\mathbf{g}_k)}{\sum_j \Phi(\mathbf{g}_j)}$$

$$g_i^{(t+1)} = \begin{cases} \neg g_i^{(t)} & \text{falls } \chi_i \leq p_{mut} \\ g_i^{(t)} & \text{sonst} \end{cases}$$

### Known Problems

- VSCode Preview: `\mbox{ tanh }` => tanh => `\text{ tanh }` => tanh
- GH Preview:
  - `\phantom{xyz}` => => ?? => ??
  - `\operatorname{tanh}` => $\operatorname{tanh}$ => `\mathop{\text{tanh}}` => tanh

$$g_i^{(t+1)} = \begin{cases} \neg \quad g_i^{(t)} & \text{falls } \chi_i \leq p_{mut} \\ \quad g_i^{(t)} & \text{sonst} \end{cases}$$

Schwierig: In Pandoc-Markdown muss Mathe mit `$` oder `$$` eingeschlossen werden, unabhängig vom konkreten Inhalt. In LaTeX ist aber `\begin{eqnarray}` bereits der Beginn einer Mathe-Umgebung, d.h. hier wären extra `$$` falsch. Das muss per Filter korrigiert werden!

```
$$\begin{eqnarray}
S &\rightarrow& a A                          \nonumber \\
A &\rightarrow& d B \ | \ b A \ | \ c A  \nonumber \\
B &\rightarrow& a c \ | \ b C \ | \ c A  \nonumber \\
C &\rightarrow& \epsilon                     \nonumber
\end{eqnarray}$$
```

should become

$$\begin{eqnarray}
S &\rightarrow& a A \\
A &\rightarrow& d B \ | \ b A \ | \ c A \\
B &\rightarrow& a c \ | \ b C \ | \ c A \\
C &\rightarrow& \epsilon
\end{eqnarray}$$

## Links

### Link to WWW

craftinginterpreters.com/the-lox-language.html

### Internal Links

selbe ebene: readme.md

unterordner: subfolder/foo.md

zurück nach oben I: ../02-parsing/antlr-parsing.md

zurück nach oben II: ../../homework/sheet01.md

## Code

```
grammar Hello;

start : stmt* ;

stmt  : ID '=' expr ';' | expr ';' ;
expr  : term ('+' term)* ;
term  : atom ('*' atom)* ;
atom  : ID | NUM ;

ID    : [a-z][a-zA-Z]* ;
NUM   : [0-9]+ ;
WS    : [ \t\n]+ -> skip ;
```

Java-Code kompilieren: `javac *.java`

**Listing 1:** The preprocessing step, cf. [@Dietz2018]

```
import org.antlr.v4.runtime.CharStreams;
import org.antlr.v4.runtime.CommonTokenStream;
import org.antlr.v4.runtime.tree.ParseTree;

public class Main {
    public static void main(String[] args) throws Exception {
        HelloLexer lexer = new HelloLexer(CharStreams.fromStream(System.in));
        CommonTokenStream tokens = new CommonTokenStream(lexer);
        HelloParser parser = new HelloParser(tokens);

        ParseTree tree = parser.start();  // Start-Regel
        System.out.println(tree.toStringTree(parser));
    }
}
```

Code ohne alles

```
import org.antlr.v4.runtime.CharStreams;
import org.antlr.v4.runtime.CommonTokenStream;
import org.antlr.v4.runtime.tree.ParseTree;

public class Main {
    public static void main(String[] args) throws Exception {
        HelloLexer lexer = new HelloLexer(CharStreams.fromStream(System.in));
        CommonTokenStream tokens = new CommonTokenStream(lexer);
        HelloParser parser = new HelloParser(tokens);

        ParseTree tree = parser.start();  // Start-Regel
        System.out.println(tree.toStringTree(parser));
    }
}
```

## Images

**Abbildung 1:** “B” (small)

**Abbildung 2:** “B”, width=“20%”

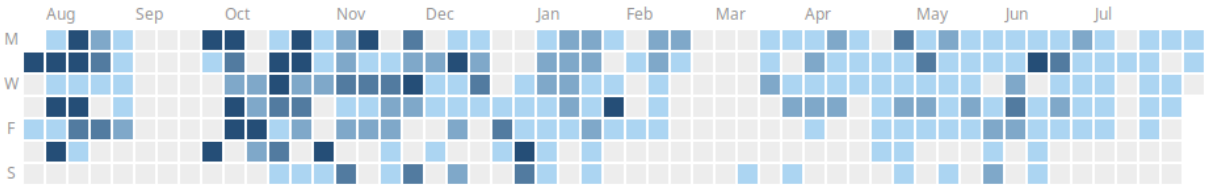


**Abbildung 3:** “wuppie” (wide)

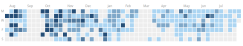

**Abbildung 4:** “wuppie”, width=“20%”

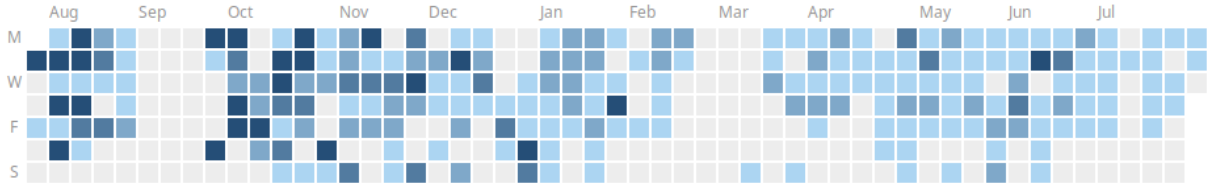


**Abbildung 5:** “wuppie” via web (raw)

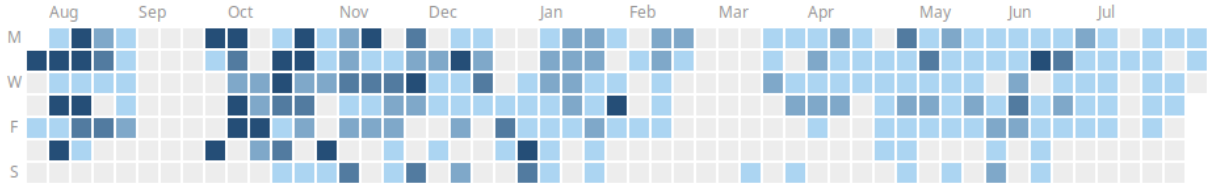


**Abbildung 6:** “wuppie” via web (raw) ( Quelle: “FooFOOOO” by me on void.intern.com )

Quelle: “Foo” by me on void.extern.com

---

**Image w/o Caption**:

ohne alles:

mit breitenangabe:

mit breitenangabe und titel:

mit caption:

**Abbildung 7:** FOO BAr Wuppie Fluppie

mit caption und breitenangabe:

**Abbildung 8:** FOO BAr Wuppie Fluppie

mit caption und breitenangabe und titel:

**Abbildung 9:** FOO BAr Wuppie Fluppie

---

**Images w/ `img`-Code**

using “5%”:

using “80px”:

mit div drumherum:

**Known Problems**

- In VSC preview as well as in LaTeX images via web like https://github.com/cagix/pandoc-thesis/blob/master/figs/wuppie.png do not work (**need to be "raw"**)

## Tabellen

mit caption:

**Tabelle 1:** Tabelle als Markdown-Pipe-Table, vgl. (Abelson, Sussmann, und Sussmann 1996)

| Rechtsbündig | Linksbündig | Default | Zentriert |
|---|---|---|---|
| foo | foo | foo | foo |
| 123 | 123 | 123 | 123 |
| bar | bar | bar | bar |

ohne caption:

| Rechtsbündig | Linksbündig | Default | Zentriert |
|---|---|---|---|
| foo | foo | foo | foo |
| 123 | 123 | 123 | 123 |
| bar | bar | bar | bar |

## Zitieren, Quellen

Normales Zitieren (Siek 2023) …

Mit Seitenangabe (Siek 2023, 111) oder Kapitel (Siek 2023, Kap. 111) …

Als Author-Zitat Siek (2023) …

## GFM

### Details

Zusammenfassung: NIX :)

Lalelu …

### Alert Extension

GH introduced “alerts” with distinctive styling, like

> [!NOTE] Foo bar, wuppie fluppie!

> [!TIP] Foo bar, wuppie fluppie!

> [!IMPORTANT] Foo bar, wuppie fluppie!

> [!WARNING] Foo bar, wuppie fluppie!

> [!CAUTION] Foo bar, wuppie fluppie!

(see https://github.blog/changelog/2023-12-14-new-markdown-extension-alerts-provide-distinctive-styling-for-significant-content/)

Let’s stick with Pandocs divs in Markdown content and use filters for export:

Foo bar, wuppie fluppie! Blablabla third line of nonsense …

Foo bar, wuppie fluppie!

Foo bar, wuppie fluppie!

Foo bar, wuppie fluppie!

Foo bar, wuppie fluppie!

- Export to GH Markdown using "distinctive alerts"
- Export to Hugo using notice shortcode
- Export to Beamer using beamercolorbox (also beameruserguide.pdf; or `block`, `alertblock`, `examples` - cf. https://www.overleaf.com/learn/latex/Beamer%23Creating_a_table_of_contents)

This should probably be in line with #180 ...

## Filter for Slides and Handouts

Foo bar, wuppie fluppie! (NOTES)

## Footnotes

Sometimes[1] we need some[2] footnotes.

## Handling of TeX Shenanigans

**Zustand:** (Formale) Beschreibung eines Zustandes der Welt

**Aktion:** (Formale) Beschreibung einer durch Agenten ausführbaren Aktion

- Anwendbar auf bestimmte Zustände
- Überführt Welt in neuen Zustand ("Nachfolge-Zustand")

LaTeX-Befehle wie `\bigskip` etc. sollten automatisch entfernt werden:

Hier nach den LaTeX-Befehlen.

**Geeignete Abstraktionen wählen für Zustände und Aktionen!**

## Credits

Typische Regeln und Konventionen tauchen überall auf, beispielsweise bei Tim Pope (siehe nächstes Beispiel) oder bei "How to Write a Git Commit Message".

```
Short (50 chars or less) summary of changes

More detailed explanatory text, if necessary.  Wrap it to about
72 characters or so.  In some contexts, the first line is treated
as the subject of an email and the rest of the text as the body.
```

[1] sometime even more often
[2] lalalala

```
The blank line separating the summary from the body is critical
(unless you omit the body entirely); tools like rebase can get
confused if you run the two together.

Further paragraphs come after blank lines.

 - Bullet points are okay, too
 - Typically a hyphen or asterisk is used for the bullet, preceded
   by a single space, with blank lines in between, but conventions
   vary here
```

Quelle: "A Note About Git Commit Messages" by Tim Pope on tbaggery.com

## Filters

### ShowMe

Hier ein ShowMe-Test:

this is hidden content …

(but not used anymore)

Use `details` instead:

this is hidden content …

**wuppie**

this is a show-me w/ title :)

### CBOX

this is content to be centered (and put into a box)…

(but not used anymore)

### Center

this content should be centered

### Alert

This will be highlighted. (but not used anymore)

Use Pandoc's mark extension instead: This will be highlighted. Even **with bold** text.

### Hinweis

This is a hint. (but not used anymore)

**Thema**

The topic of this task or … (but not used anymore)

**BSP**

Lalalelu

Simple Beispiel-Button X (but not used anymore)

lalelu

use `ex` instead:

Simple Beispiel-Button X

Beispiel-Button w/ link (https://github.com/sdiehl/write-you-a-haskell/blob/master/README.md)

Vor `\pause`…

Nach `\pause`…

## Zum Nachlesen

- Tate (2010, Kap. 2)
- Nystrom (2021)

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**Lernziele**

- k1: K1
- k2: K2
- k3: K3.1
- k3: K3.2

**Quizzes**

- Quiz Git Basics (ILIAS)

**Challenges**

**Lexer und Parser mit ANTLR: Programmiersprache Lox**

Betrachten Sie folgenden Code-Schnipsel in der Sprache "Lox":

```
fun fib(x) {
    if (x == 0) {
        return 0;
    } else {
        if (x == 1) {
            return 1;
        } else {
            fib(x - 1) + fib(x - 2);
        }
    }
}

var wuppie = fib(4);
```

Erstellen Sie für diese fiktive Sprache einen Lexer+Parser mit ANTLR. Implementieren Sie mit Hilfe des Parse-Trees und der Listener oder Visitoren einen einfachen Pretty-Printer.
(Die genauere Sprachdefinition finden Sie bei Bedarf unter craftinginterpreters.com/the-lox-language.html.)

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**Quellen**


Abelson, H., G. J. Sussmann, und J. Sussmann. 1996. *Structure and Interpretation of Computer Programs*. MIT Press. https://mitpress.mit.edu/sites/default/files/sicp/index.html.

Nystrom, R. 2021. *Crafting Interpreters*. Genever Benning. https://github.com/munificent/craftinginterpreters.

Parr, T. 2014. *The Definitive ANTLR 4 Reference*. Pragmatic Bookshelf. https://learning.oreilly.com/library/view/the-definitive-antlr/9781941222621/.

Siek, J. G. 2023. *Essentials of Compilation: An Incremental Approach in Racket*. The MIT Press. https://github.com/IUCompilerCourse/Essentials-of-Compilation.

Tate, B. A. 2010. *Seven Languages in Seven Weeks*. Pragmatic Bookshelf. https://learning.oreilly.com/library/view/seven-languages-in/9781680500059/.

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**Exceptions:**

- “FooFOOOO” by me on void.intern.com
- “A Note About Git Commit Messages” by Tim Pope on tbaggery.com
- “Foo” by me on void.extern.com

> **Last modified:** 3b33179 (extend tests (image skaling), 2025-04-09)